

Lemma (整数环无零因子) 对 $\forall x, y \in \mathbb{Z}$. 若 $x \neq 0$ 且 $y \neq 0$, 则有: $xy \neq 0$.

Proof: 假设 $xy = 0$. 则有: $xy = 0 = 0y$

$\therefore x, 0, y \in \mathbb{Z}$, $xy = 0y$ 且 $y \neq 0 \quad \therefore x = 0$ 与 $x \neq 0$ 矛盾.

$\therefore xy \neq 0 \quad \square$

Lemma: 对 $\forall x, y \in \mathbb{Z}$, 若 $xy = 0$, 则有: $x = 0$ 或 $y = 0$.

Proof: 这就是上一引理的逆否命题. $\square$

Lemma: $(\mathbb{Z}, +, \cdot, 0, 1)$ 是一个整环.

Proof: $\because (\mathbb{Z}, +, \cdot, 0, 1)$ 是一个非零交换环, 且无零因子.

$\therefore (\mathbb{Z}, +, \cdot, 0, 1)$ 是一个整环. $\square$

Lemma: 对 $\forall [r, s] \in \mathbb{Q}$, $\forall \alpha \in \mathbb{Z} \setminus \{0\}$, 有: $[r, s] = [r\alpha, s\alpha]$

Proof: $\because [r, s] \in \mathbb{Q} \quad \therefore r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}$.

$\because r \in \mathbb{Z}, \alpha \in \mathbb{Z} \setminus \{0\} \quad \therefore r\alpha \in \mathbb{Z}$

$\because s \in \mathbb{Z} \setminus \{0\}, \alpha \in \mathbb{Z} \setminus \{0\} \quad \therefore s\alpha \in \mathbb{Z} \setminus \{0\} \quad \therefore [r\alpha, s\alpha] \in \mathbb{Q}$.

$\because r(s\alpha) = r(\alpha s) = (r\alpha)s \quad \therefore (r, s) \sim (r\alpha, s\alpha)$

$\therefore [r, s] = [r\alpha, s\alpha] \quad \square$

有理数集 $\mathbb{Q}$ 的构造

Lemma: 定义集合 $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ 上的二元关系为: ~~$\forall (r, s), (r', s') \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$~~

$\forall (r, s), (r', s') \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$,

$$(r, s) \sim (r', s') \Leftrightarrow rs' = r's$$

则有: $(r, s) \sim (r', s')$ 是集合 $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ 上的等价关系.

Proof: $\forall (r, s) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, $\therefore rs = rs$

$\therefore (r, s) \sim (r, s)$ $\therefore$ 反身性成立.

$\forall (r, s), (r', s') \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, 若 $(r, s) \sim (r', s')$, 则有:

$$\therefore (r, s) \sim (r', s') \quad \therefore rs' = r's \quad \therefore r's = rs'$$

$\therefore (r', s') \sim (r, s)$ $\therefore$ 对称性成立.

$\forall (r, s), (r', s'), (r'', s'') \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, 若 $(r, s) \sim (r', s')$ 且 $(r', s') \sim (r'', s'')$,

$$\text{则有: } \therefore (r, s) \sim (r', s') \quad \therefore rs' = r's$$

$$\therefore (r', s') \sim (r'', s'') \quad \therefore r's'' = r''s'$$

$$\therefore (rs')(r's'') = (r's)(r''s')$$

$$\therefore (rs'')(r's') = (r''s)(r's') \quad \therefore (rs'')r' = (r''s)r'$$

$$\text{若 } r' = 0, \text{ 则有: } r = 0 \text{ 且 } r'' = 0 \quad \therefore rs'' = r''s$$

$$\text{若 } r' \neq 0, \text{ 则有: } rs'' = r''s$$

$$\therefore rs'' = r''s \quad \therefore (r, s) \sim (r'', s'') \quad \therefore \text{传递性成立.}$$



定义(有理数集 $\mathbb{Q}$) 定义有理数集 $\mathbb{Q}$ 为 $\mathbb{Q} := \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) / \sim$

记 $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ 中包含 (r, s) 的等价类为 $[r, s]$.

Remark: 对 $\forall (r, s) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ (其中 $r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}$), 有:

$$[r, s] \subseteq \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$$

$$[r, s] = \{(x, y) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) \mid (x, y) \sim (r, s)\}$$

$$= \{(x, y) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) \mid xs = ry\}$$

$$\mathbb{Q} := \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) / \sim = \{[r, s] \mid (r, s) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})\}$$

Lemma (从 $\mathbb{Z}$ 到 $\mathbb{Q}$ 的嵌入映射) 定义映射 $f: \mathbb{Z} \rightarrow \mathbb{Q}$ 为:

$$f: \mathbb{Z} \longrightarrow \mathbb{Q}$$

$$x \longmapsto [x, 1]$$

则有: 映射 f 是单射.

proof: 对 $\forall x \in \mathbb{Z}$ $\because x \in \mathbb{Z}, 1 \in \mathbb{Z} \setminus \{0\} \therefore (x, 1) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$

$$\therefore [x, 1] \in \mathbb{Q} \quad \therefore f(x) = [x, 1] \in \mathbb{Q} \quad \therefore f(\mathbb{Z}) \subseteq \mathbb{Q}$$

对 $\forall x_1, x_2 \in \mathbb{Z}$,

$$\text{若 } x_1 = x_2, \text{ 则有: } (x_1, 1) = (x_2, 1) \quad \therefore [x_1, 1] = [x_2, 1]$$

$$\therefore f(x_1) = [x_1, 1] = [x_2, 1] = f(x_2) \quad \therefore f \text{ 是映射.}$$

若 $f(x_1) = f(x_2)$, 则有: $[x_1, 1] = [x_2, 1] \quad \therefore (x_1, 1) \sim (x_2, 1)$

$\therefore x_1 \cdot 1 = x_2 \cdot 1 \quad \therefore x_1 = x_2 \quad \therefore f$ 是单射. $\square$

Remark: 今后将不加说明地将 $\mathbb{Z}$ 视同 $\mathbb{Q}$ 的子集

定义 ($\mathbb{Q}$ 中的加法和乘法) 对 $\forall [r_1, s_1], [r_2, s_2] \in \mathbb{Q}$, 定义它们的和与积为:

$$[r_1, s_1] + [r_2, s_2] := [r_1 s_2 + r_2 s_1, s_1 s_2]$$

$$[r_1, s_1] \cdot [r_2, s_2] := [r_1 r_2, s_1 s_2]$$

Lemma: 对 $\forall s_1, s_2 \in \mathbb{Z} \setminus \{0\}$, 有: $[0, s_1] = [0, s_2]$

Proof: $\because 0 \cdot s_2 = 0 = 0 \cdot s_1 \quad \therefore (0, s_1) \sim (0, s_2)$

$$\therefore [0, s_1] = [0, s_2] \quad \square$$

定义 ($\mathbb{Q}$ 中的加法零元) 任取 $s \in \mathbb{Z} \setminus \{0\}$. 定义 $0 := [0, s]$

Remark: 上引理保证了这一定义的良好性.

Lemma ($\mathbb{Q}$ 中 0 的加法性质) 对 $\forall x \in \mathbb{Q}$, 有: $x + 0 = x = 0 + x$.

Proof: $\because x \in \mathbb{Q} \quad \therefore \exists r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } x = [r, s]$

任取 $\alpha \in \mathbb{Z} \setminus \{0\} \quad \therefore 0 = [0, \alpha]$

$$\therefore x + 0 = [r, s] + [0, \alpha] = [r\alpha + 0s, s\alpha] = [r\alpha, s\alpha] = [r, s] = x$$

$$0 + x = [0, \alpha] + [r, s] = [0s + r\alpha, \alpha s] = [r\alpha, s\alpha] = [r, s] = x$$

$$\therefore x + 0 = x = 0 + x \quad \square$$

Lemma ($\mathbb{Q}$ 中0的乘法性质) 对 $\forall x \in \mathbb{Q}$, 有: $x \cdot 0 = 0 = 0 \cdot x$

Proof: $\because x \in \mathbb{Q} \quad \therefore \exists r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } x = [r, s]$

任取 $\alpha \in \mathbb{Z} \setminus \{0\} \quad \therefore 0 = [0, \alpha]$

$$\therefore x \cdot 0 = [r, s] \cdot [0, \alpha] = [r \cdot 0, s\alpha] = [0, s\alpha] = 0$$

$$0 \cdot x = [0, \alpha] \cdot [r, s] = [0 \cdot r, \alpha s] = [0, \alpha s] = 0$$

$$\therefore x \cdot 0 = 0 = 0 \cdot x \quad \square$$

Lemma: $\lambda \in \mathbb{Q}$ 若对 $\forall x \in \mathbb{Q}$, 有: $x + \lambda = x = \lambda + x$, 则有: $\lambda = 0$.

Proof: $\because \lambda \in \mathbb{Q} \quad \therefore \exists \gamma \in \mathbb{Z}, \delta \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } \lambda = [\gamma, \delta]$

$\because x \in \mathbb{Q} \quad \therefore \exists r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } x = [r, s]$

$$\because x + \lambda = x \quad \therefore [r, s] + [\gamma, \delta] = [r, s]$$

$$\therefore [r\delta + \gamma s, s\delta] = [r, s] \quad \therefore (r\delta + \gamma s, s\delta) \sim (r, s)$$

$$\therefore (r\delta + \gamma s)s = r(s\delta) \quad \therefore r s \delta + \gamma s^2 = r s \delta$$

$$\therefore \gamma s^2 + r s \delta = 0 + r s \delta \quad \therefore \gamma s^2 = 0 \quad \therefore \gamma = 0 \text{ 或 } s^2 = 0$$

$$\because s \in \mathbb{Z} \setminus \{0\} \quad \therefore s^2 \in \mathbb{Z} \setminus \{0\} \quad \therefore \gamma = 0$$

$$\therefore \lambda = [\gamma, \delta] = [0, \delta] = 0 \quad \square$$